

Practical Tasks

Task-1

Develop a C program using fork() that creates a parent and child process. Display the Process ID (PID), Parent Process ID (PPID), and process states at different stages of execution

```
[vigneshkumar@vigneshs-MacBook-Air ~ % nano fork.c
```

Code

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <stdlib.h>

int main() {
    pid_t pid;

    printf("Before fork():\n");
    printf("Process PID: %d\n", getpid());
    printf("Parent PID: %d\n\n", getppid());

    pid = fork();

    if (pid < 0) {
        perror("fork failed");
        return 1;
    }
    if (pid == 0) {
        printf("CHILD PROCESS\n");
        printf("PID : %d\n", getpid());
        printf("PPID : %d\n", getppid());
        printf("State: Running\n");

        sleep(5);

        printf("\nChild process after sleep\n");
        printf("PID : %d\n", getpid());
        printf("PPID : %d\n", getppid());
        printf("State: Running\n");

        printf("Child process terminating...\n");
        exit(0);
    }
    else {
        printf("PARENT PROCESS\n");
        printf("PID : %d\n", getpid());
        printf("PPID : %d\n", getppid());
        printf("Child PID: %d\n", pid);
        printf("State: Running\n");

        printf("\nParent waiting for child...\n");
        wait(NULL);

        printf("Child process completed.\n");
        printf("Parent process terminating...\n");
    }

    return 0;
}
```

```
[vigneshkumar@vigneshs-MacBook-Air ~ % nano fork.c  
[vigneshkumar@vigneshs-MacBook-Air ~ % gcc fork.c -o fork  
[vigneshkumar@vigneshs-MacBook-Air ~ % ./fork
```

Output

```
[vigneshkumar@vigneshs-MacBook-Air ~ % nano fork.c  
[vigneshkumar@vigneshs-MacBook-Air ~ % gcc fork.c -o fork  
[vigneshkumar@vigneshs-MacBook-Air ~ % ./fork  
Before fork():  
Process PID: 5218  
Parent PID: 5131  
  
PARENT PROCESS  
PID : 5218  
PPID : 5131  
Child PID: 5221  
State: Running  
  
Parent waiting for child...  
CHILD PROCESS  
PID : 5221  
PPID : 5218  
State: Running  
  
Child process after sleep  
PID : 5221  
PPID : 5218  
State: Running  
Child process terminating...  
Child process completed.  
Parent process terminating...  
vigneshkumar@vigneshs-MacBook-Air ~ %
```

Task-2

Design an experiment to observe process state transitions (Ready, Running, Waiting, Terminated) using Linux monitoring tools such as ps, top, and /proc. Document the observations.

```
vigneshkumar@vigneshs-MacBook-Air ~ % nano Task-p2.c
vigneshkumar@vigneshs-MacBook-Air ~ %
```

Code

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <stdlib.h>
#include <time.h>

int main() {
    pid_t pid = fork();

    if (pid < 0) {
        perror("fork failed");
        return 1;
    }

    if (pid == 0) {
        printf("Child Process Started\n");
        printf("PID : %d\n", getpid());
        printf("PPID : %d\n", getppid());

        printf("\nState 1: WAITING/SLEEPING\n");
        sleep(10);

        printf("\nState 2: RUNNING\n");

        time_t start = time(NULL);

        while (time(NULL) - start < 15) {
            volatile unsigned long long x = 0;
            for (unsigned long long i = 0; i < 10000000; i++) {
                x += i;
            }
        }

        printf("\nState 3: TERMINATING\n");
        exit(0);
    }
    else {
        printf("Parent Process\n");
        printf("PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);

        printf("\nParent is waiting for 35 seconds.\n");
        sleep(35);

        printf("\nParent calling wait().\n");
        wait(NULL);

        printf("Child process collected.\n");
        printf("Parent terminating.\n");
    }

    return 0;
}
```

```
vigneshkumar@vigneshs-MacBook-Air ~ % nano Task-p2.c
vigneshkumar@vigneshs-MacBook-Air ~ % gcc Task-p2.c -o Task-p2
vigneshkumar@vigneshs-MacBook-Air ~ % ./Task-p2
```

OUTPUT

```
[vigneshkumar@vigneshs-MacBook-Air ~ % nano Task-p2.c
[vigneshkumar@vigneshs-MacBook-Air ~ % gcc Task-p2.c -o Task-p2
[vigneshkumar@vigneshs-MacBook-Air ~ % ./Task-p2
Parent Process
PID      : 5258
Child PID : 5259
Child Process Started

Parent is waiting for 35 seconds.
PID : 5259
PPID : 5258

State 1: WAITING/SLEEPING

State 2: RUNNING

State 3: TERMINATING

Parent calling wait().
Child process collected.
Parent terminating.
vigneshkumar@vigneshs-MacBook-Air ~ %
```